Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology
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WEEKLY LESSON PLAN

Department: Computer Science and Engineering

Year / Semester: 2025-26 / Winter

Course Code / Course Name: 10213CS224 / Full Stack Application Development

Unit IV: Version Control, Build Tools and DevOps Foundations **Course Outcome: Apply version control, build tools, and CI/CD practices for application delivery.**				
Week	**Session**	**Topics to be Covered**	**Problems to Practice (Hands-On)**	**Resources**
Week 11	Session 51	Introduction to Jenkins & CI/CD	**Task 11.1:** Install Jenkins on local system / VM.2. Unlock Jenkins and configure admin user.3. Install recommended plugins.4. Create first Freestyle job.5. Execute a simple build and view console output.	https://youtu.be/FX322RVNGj4
	Session 52	Jenkins + Git Integration	**Task 11.2:** Connect Jenkins with GitHub repository.2. Configure SCM in Jenkins job.3. Pull source code from GitHub.4. Trigger build manually after commit.5. Configure GitHub webhook for auto build.	https://www.jenkins.io/doc/tutorials/
	Session 53	Jenkins Pipeline Basics	**Task 11.3:** Create a Jenkins Pipeline job.2. Write a basic Jenkinsfile.3. Create stages: Checkout, Build, Test.4. Store Jenkinsfile in GitHub.5. Execute pipeline from SCM.	https://www.jenkins.io/doc/book/pipeline/
	Session 54	Automated Build & Testing	**Task 11.4:** Integrate unit tests in Jenkins pipeline.2. Fail build on test failure.3. Archive test results.4. Generate build artifacts.5. Archive	https://youtu.be/7KCS70sCoK0

			artifacts in Jenkins.	
	Session 55	CI/CD Deployment & Cloud Integration	**Task 11.5:** Deploy application to staging server.2. Add manual approval step for production.3. Send email/Slack notifications.4. Use parameters for environment selection.5. Monitor pipeline execution and logs.	https://www.jenkins.io/doc/book/using/

Task 11.1 Solution: Introduction to Jenkins & CI/CD

1. Install Jenkins

- Install Java (JDK 11 or 17)
- Download Jenkins WAR or install via package manager
- Start Jenkins using:

```
java -jar jenkins.war
```

2. Unlock Jenkins

- Access Jenkins at `http://localhost:8080`
- Enter the initial admin password from:

```
/var/lib/jenkins/secrets/initialAdminPassword
```

3. Install Plugins

- Select **Install suggested plugins**
- Verify installation of:
  - Git

    - Pipeline
    - Credentials Binding

4. Create First Freestyle Job

- Click **New Item → Freestyle Project**
- Add **Build Step → Execute Shell**

echo "Hello Jenkins"

5. Build & Verify

- Click **Build Now**
- Check **Console Output**
- Output displays:
  ✓ *Hello Jenkins*

Task 11.2 Solution: Jenkins + Git Integration

1. Create GitHub Repository

- Create a repository (example: `jenkins-demo`)
- Add a sample file (`index.html` or `README.md`)

2. Configure Jenkins Job

- Open Jenkins Job → **Configure**
- Under **Source Code Management**
    - Select **Git**
    - Enter GitHub repository URL

3. Build Job

- Click **Build Now**
- Verify code is pulled in **Workspace**

4. Configure GitHub Webhook

- In GitHub repo → **Settings** → **Webhooks**
- Payload URL:

```
http://<jenkins-ip>:8080/github-webhook/
```

5. Verify Auto Trigger

- Commit code to GitHub
- Jenkins build starts automatically

Task 11.3 Solution: Jenkins Pipeline Basics

1. Create Pipeline Job

- Jenkins → **New Item** → **Pipeline**

2. Jenkinsfile (Basic Pipeline)

Create a `Jenkinsfile` in GitHub repo:

```
pipeline {
    agent any

    stages {
        stage('Checkout') {
            steps {
```

```
                git 'https://github.com/user/jenkins-demo.git'
            }
        }

        stage('Build') {
            steps {
                echo 'Building the application'
            }
        }

        stage('Test') {
            steps {
                echo 'Running tests'
            }
        }
    }
}
```

3. Configure Pipeline

- Select **Pipeline script from SCM**
- Choose **Git**
- Enter repo URL

4. Run Pipeline

- Click **Build Now**
- Verify all stages execute successfully

Task 11.4 Solution: Automated Build & Testing

1. Add Test Stage

Update Jenkinsfile:

```
stage('Test') {
    steps {
        sh 'echo "Running unit tests"'
    }
}
```

2. Fail Build on Test Failure

```
stage('Test') {
    steps {
        sh 'exit 1'
    }
}
```

Build fails when test fails

3. Archive Artifacts

```
stage('Package') {
    steps {
        sh 'mkdir output && echo "Build file" > output/app.txt'
        archiveArtifacts artifacts: 'output/*'
    }
}
```

4. Verify

- Artifacts appear under **Build → Artifacts**
- Test results visible in logs

Task 11.5 Solution: CI/CD Deployment & Cloud Integration

1. Parameterized Pipeline

```
parameters {
    choice(name: 'ENV', choices: ['DEV', 'QA', 'PROD'], description: 'Select Environment')
}
```

2. Deployment Stage

```
stage('Deploy') {
    steps {
        echo "Deploying to ${params.ENV} environment"
    }
}
```

3. Manual Approval for Production

```
stage('Approval') {
    when {
        expression { params.ENV == 'PROD' }
    }
    steps {
        input message: 'Approve Production Deployment?'
    }
}
```

4. Notifications

```
post {
    success {
        echo 'Build Successful'
    }
    failure {
        echo 'Build Failed'
    }
}
```

5. Monitoring

- View pipeline stages
- Analyze logs
- Track execution time

CI/CD pipeline completed successfully.